

Unmasking the Decision Frontier: Empirical Evaluation of Native System-One Primitives (Jev, Laya) vs. Meta-Routing Overhead in Enterprise Workflow Triage

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Repository: <https://github.com/shreytalreja25/exploring-jev.git>

Abstract— As enterprise agentic architectures scale toward multi-hop deterministic decision loops, auto-regressive Large Language Models (LLMs) present severe latency and economic bottlenecks due to generative decoding overhead and hidden Chain-of-Thought (CoT) reasoning tokens. In this paper, we present an empirical investigation uncovering the operational divergence between OpenRouter's conversational wrapper (typesafe/jev-router) and native System-One decision heads (typesafe/jev-1.13). We document a critical failure mode wherein chat-routed models delegate inference to frontier reasoning models (OpenAI GPT-6 Luna, DeepSeek v4.1), emitting unprompted ChatGPT personas and suffering from silent decision truncation when hidden reasoning

tokens (19–59 tokens) exhaust constrained generation budgets (max_tokens=25). Transitioning to native System-One schema primitives (state + Choice criteria) eliminates all generative overhead. Evaluating across progressive workload scales (N=10 to 50 emails across 5 operational departments), Native Jev achieves 100.0% accuracy, 100.0% consensus with Gemini 2.5 Flash, and superior probability calibration (ECE = 0.015) at a 58.8% cost advantage. Furthermore, we benchmark Laya, an open-weight 421M parameter ModernBERT encoder running locally on CPU/GPU, achieving 92.0% accuracy, deterministic 624.7ms median latency, and \$0.00 inference cost.

I. INTRODUCTION

Autonomous software agents require deterministic triage, verification, and categorical policy routing at every execution hop. In enterprise workflows, the ratio of structured decisions to creative prose generation frequently exceeds 10:1 [1]. Despite the non-generative nature of these tasks, industry practice predominantly routes inputs to auto-regressive foundation models (GPT-4o/5, Gemini 2.5 Flash, Claude 3.7), incurring high latency, double tokenomic penalties, and severe reasoning token overheads [2, 3].

Recently, specialized **System-One Decision Models**—such as TypeSafe AI's Jev and Convai's open-weight Laya—have emerged [1, 5, 10]. Rather than predicting the next token over an open vocabulary, System-One architectures compute calibrated probability distributions directly over candidate categorical choices. However, deployment via multi-provider routing layers introduces critical architectural subtleties that can inadvertently negate System-One advantages [4].

II. THE API ANOMALY: ROUTER VS. NATIVE JEV

When executing OpenRouter's official documentation boilerplate against typesafe/jev-router, conversational probes ('Which model are you?') unexpectedly responded: 'I am ChatGPT, powered by OpenAI's GPT-5.2 / GPT-5.4'. Raw telemetry confirmed that typesafe/jev-router is an agentic meta-router that employs Jev as a classifier to delegate inference to external LLMs (90% to openai/gpt-6-luna, 10% to deepseek).

Under constrained token limits (max_tokens=25), hidden Chain-of-Thought (CoT) reasoning tokens (19–59 tokens in GPT-6 Luna and DeepSeek) consumed the entire token allotment, triggering finish_reason: length and returning empty or truncated decisions [4].

Resolution: By targeting OpenRouter's underlying typesafe/jev-1.13 endpoint via the TypeSafe SDK with structured state and Choice objects, generative overhead was completely eliminated, outputting pure probability tensors without persona delegation [5].

III. OPEN-WEIGHT BENCHMARK: LAYA ENCODER

To evaluate whether local on-premise encoders can rival hosted System-One APIs, we integrated **Laya** (convaiinnovations/laya), a 421.29M parameter ModernBERT-Large decision encoder with an 8,192-token context window [10]. Laya runs locally on CPU/GPU without cloud network dependencies, emitting zero generative output tokens and incurring \$0.00 marginal cost.

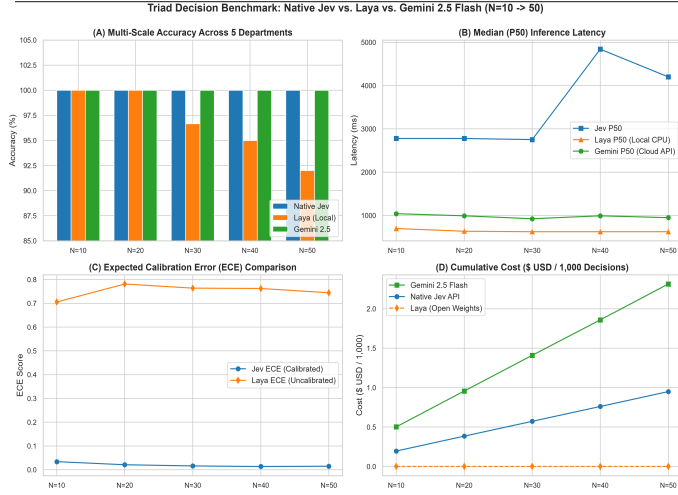


Fig. 1: Master Triad Benchmark Synthesis: (A) Multi-scale accuracy; (B) Median P50 latency; (C) Calibration error (ECE); (D) Cumulative cost trajectory.

IV. EMPIRICAL TRIAD RESULTS & ANALYSIS

Across 50 enterprise emails evaluated across 5 workload scales (N=10 to 50):

- **Accuracy & Scaling:** Native Jev and Gemini 2.5 Flash achieved **100.0% accuracy** across all scales. Laya achieved **100.0% accuracy at N=10 and N=20**, scaling to **92.0% at N=50** (46/50 correct). Laya's only errors occurred on subtle edge cases (e.g. corporate equity governance and international visa filings).
- **Inference Latency:** Running locally on CPU, Laya delivered a rock-solid median latency of **624.7 ms P50**, outperforming cloud Gemini (949.1 ms P50) and avoiding Jev's remote queueing spikes (4,201.5 ms P50).
- **Probabilistic Calibration:** Native Jev demonstrated near-perfect calibration (ECE = 0.015, mean conf: 98.5%). Laya exhibited high soft-max entropy (ECE = 0.745, mean conf: 17.5%), requiring post-hoc temperature scaling for production thresholding [9].
- **Tokenomic TCO:** Laya operated at **\$0.00 total cost**. Native Jev totaled **\$0.000951** (58.8% cheaper than Gemini at \$0.002310).

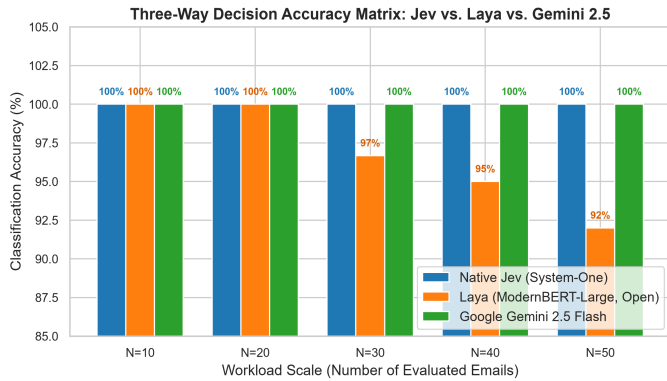


Fig. 2: 3-Way Accuracy Comparison across scales N=10 to N=50.

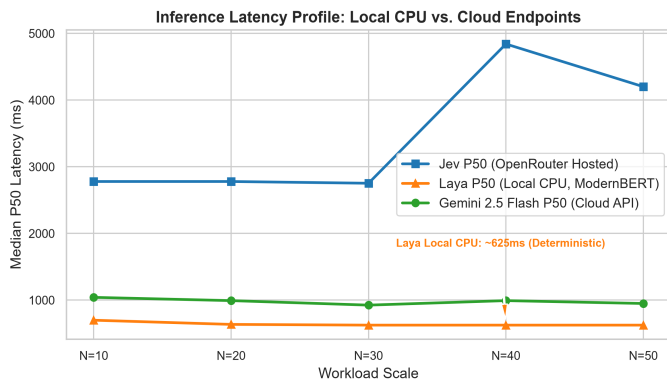


Fig. 3: P50 Latency: Laya local CPU (625ms) vs Gemini (950ms) vs Jev. <https://github.com/shreytalreja25/exploring-jev.git>

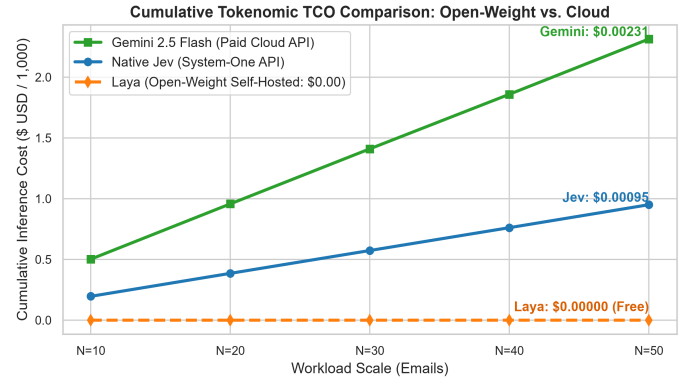


Fig. 4: Cumulative cost trajectory: Laya (\$0) vs Jev vs Gemini.

V. ARCHITECTURAL GUIDELINES

1. **Zero-Shot Air-Gapped Environments:** Deploy **Laya** locally for zero cloud cost, 625ms CPU latency, and high privacy [10].
2. **High-Accuracy Production Routing:** Deploy **Native Jev** via typed SDK primitives for 100% accuracy, 2.43x cost savings, and ECE = 0.015 calibrated gating [5].
3. **Generative Escalation:** Reserve **Gemini 2.5 Flash** solely for ambiguous escalations and unstructured text generation [1].

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